

Behzad Ousat

Applied Scientist | Security ML | Ph.D. Candidate

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📍 Miami, FL - Open to relocate

Ph.D. researcher in applied machine learning for web security, with first-author papers at ACM WWW, RAID, and DIMVA on bot detection, Captcha security, and LLM-based agents. Builds the full research stack end to end: web-scale measurement pipelines, model training and evaluation, and adversarial robustness analysis. Backed by 4 years of production infrastructure engineering.

EDUCATION

Ph.D. Computer Science, Florida International University	01/2022 – 12/2026
M.Sc. Computer Engineering, Sharif University of Technology	08/2018 – 08/2021
B.Sc. Computer Engineering, University of Tehran	08/2014 – 08/2018

RESEARCH EXPERIENCE

Florida International University - Research Assistant, Applied ML | Miami, FL 01/2022 – Present

LLM Agents and Automated Evaluation

- Developed **LLM-powered browser agents** that mimic human interaction patterns, reaching 90%+ success on real-world web crawling tasks, then used them to re-evaluate commercial bot defenses.
- Benchmarked 10 LLMs for automated vulnerability analysis in **CI/CD** pipelines, building an evaluation harness that measured precision, recall, and latency on patched and vulnerable packages (RAID '26).

Modeling and Adversarial Robustness

- Trained **LSTM** and **Transformer** models in **PyTorch** on browser interaction time series to classify automated agents at 95% accuracy, ablating each input modality to isolate its contribution.
- Attacked behavioral bot detectors with **adversarial ML** (**FGSM**, genetic algorithms) to quantify the practical limits of multi-modal defenses (WWW '24).
- Built distributed training and inference workflows over terabytes of **PCAP** data, applying **open set recognition** to detect zero-day network attack classes (JISA '23).

Web-Scale Measurement Pipelines

- Operated **multi-modal** collection pipelines that captured 1.6B+ browser interaction artifacts (DOM, screenshots, event streams) into terabytes of **NoSQL** storage.
- Studied development practices and code reuse across **300K+ phishing websites** via a **dockerized** pipeline that produced 100+ GB of artifacts; joint work with **Microsoft Security** (DIMVA '24).
- Measured Captcha deployment across 1M+ websites, then built an **object detection** pipeline that solved 12 schemes at 80%+ accuracy (WWW '24).

Fraunhofer IDMT - Machine Learning Engineer Intern | Germany 09/2019 – 12/2019

- Built a **face recognition** pipeline (OpenCV, **CNN**) to identify studio scenes across 100+ hours of news video.

INDUSTRY EXPERIENCE

Tapsi - Software and DevOps Engineer | Tehran 12/2017 – 10/2021

Platform Technical Team Lead

- Created **internal platform tooling** (Python, Bash) that let developers run a single microservice locally against shared **Kubernetes** staging environments, cutting local resource needs by 80%.
- Developed **Node.js** and Python message-passing libraries over **gRPC** and **RabbitMQ** that sustained 10K events per second, instrumented with **Prometheus** and **Grafana**.

Senior Infrastructure Engineer

- Migrated 60+ microservices to **Kubernetes**, coordinating with backend teams and onboarding developers through internal workshops.
- Engineered production data pipelines with **Kafka** and **Apache Spark** to automate collection, processing, and reporting for Python ML microservices and analytics workflows.

LLM-Powered Chat-with-PDF System - Startup MVP

06/2025 – 12/2025

- Built a **Chat-with-PDF** system across 30K+ clinical trial documents, combining **RAG** and **prompt chaining** on **Google Cloud** APIs with a **Python** ingestion and embedding pipeline on AWS S3.

SKILLS

Languages: Python | Node.js | Java | C/C++ | Bash

ML & Modeling: PyTorch | TensorFlow | Transformers | scikit-learn | Pandas | NumPy

Research Methods: LLM Agents | RAG | Adversarial ML | Open Set Recognition | Time Series | LLM Evaluation

Measurement: Web Crawling at Scale | Bot & Phishing Detection | PCAP Analysis | Docker

Data & Infra: PostgreSQL | MongoDB | Kafka | PySpark | Kubernetes | Terraform | AWS | GCP

TEACHING, MENTORING, AND SERVICE

Teaching Assistant , Software Security, Operating Systems, Python Programming, FIU	2021–2025
Graduate Mentor , Research methodology, coding practices, and project planning	2022–2025
Journal Reviewer , IEEE Transactions on Information Forensics & Security (TIFS, IF 8.0)	2024
Conference Reviewer , RAID, DIMVA, MadWeb	2024, 2025
Dissertation Year Fellowship Award , Florida International University	2026

PUBLICATIONS

B. Ousat, L. Rampersaud, D. Bailey, N. Turkmen, A. Kharraz, Broken Gates: Re-evaluating Web Bot Defenses in the Age of LLM Agents. Preprint, 2026. arXiv:2607.18659

S. A. Akhavani, **B. Ousat**, A. Kharraz, Vulnerability Evolution and the Promise of Automated Gatekeeping in Open-Source Software. International Symposium on Research in Attacks, Intrusions and Defenses (RAID), 2026.

B. Ousat, M. Shariatnasab, E. Schafir, F. Shirani, A. Kharraz, In-Application Defense Against Evasive Web Scans through Behavioral Analysis. Preprint, 2025. arXiv:2412.07005

B. Ousat, D. Luo, A. Kharraz, Breaking the Bot Barrier: Evaluating Adversarial AI Against Multi-Modal Defenses. ACM Web Conference (WWW), 2024.

B. Ousat, E. Schafir, D. C. Hoang, M. A. Tofighi, S. Arshad, C. Nguyen, S. Uluagac, A. Kharraz, The Matter of Captchas: An Analysis of a Brittle Security Feature on the Modern Web. ACM Web Conference (WWW), 2024.

M. A. Tofighi, **B. Ousat**, J. Zandi, E. Schafir, A. Kharraz, Constructs of Deceit: Exploring Nuances in Modern Social Engineering Attacks. Conference on Detection of Intrusions and Malware & Vulnerability Assessment (DIMVA), 2024.

B. Ousat, M. A. Tofighi, A. Kharraz, An End-to-End Analysis of Covid-Themed Scams in the Wild. ACM ASIA Conference on Computer and Communications Security (ASIACCS), 2023.

M. Soltani, **B. Ousat**, M. Jafari Siavoshani, A. H. Jahangir, An Adaptable Deep Learning-based Intrusion Detection System to Zero-day Attacks. Journal of Information Security and Applications, 2023.

L. Rampersaud, **B. Ousat**, S. A. Akhavani, J. Zandi, S. Uluagac, A. Kharraz, Evaluating Security Checks Against Malicious Payloads with Forged Signatures. IEEE IEMCON, 2025.

L. Rampersaud, **B. Ousat**, C. Brown, S. Uluagac, A. Kharraz, On the Effectiveness of End-Users' Data Backup Practices Against Data Corruption. IEEE IEMCON, 2025.